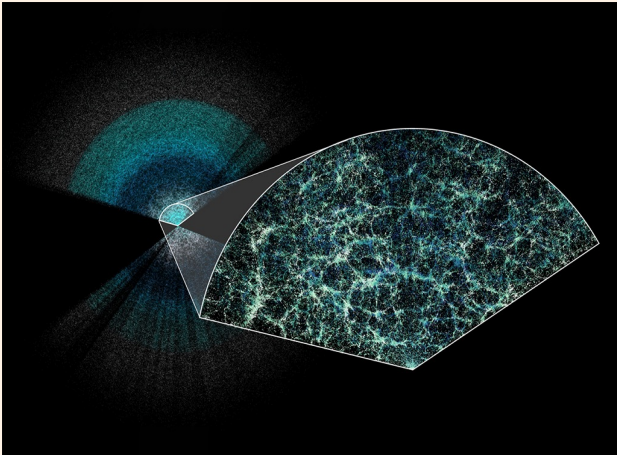


# WILL $\Lambda$ CDM SURVIVE? PERSPECTIVES FROM DES AND DESI

Ofer Lahav

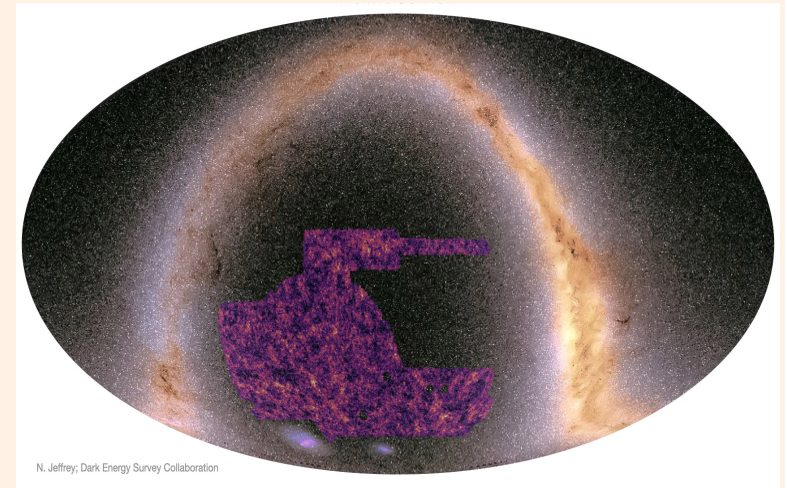
University College London

(and a member of DES & DESI)

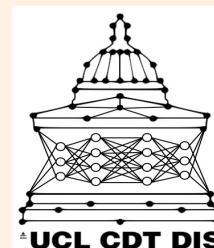


DESI galaxy map

Andrew@60



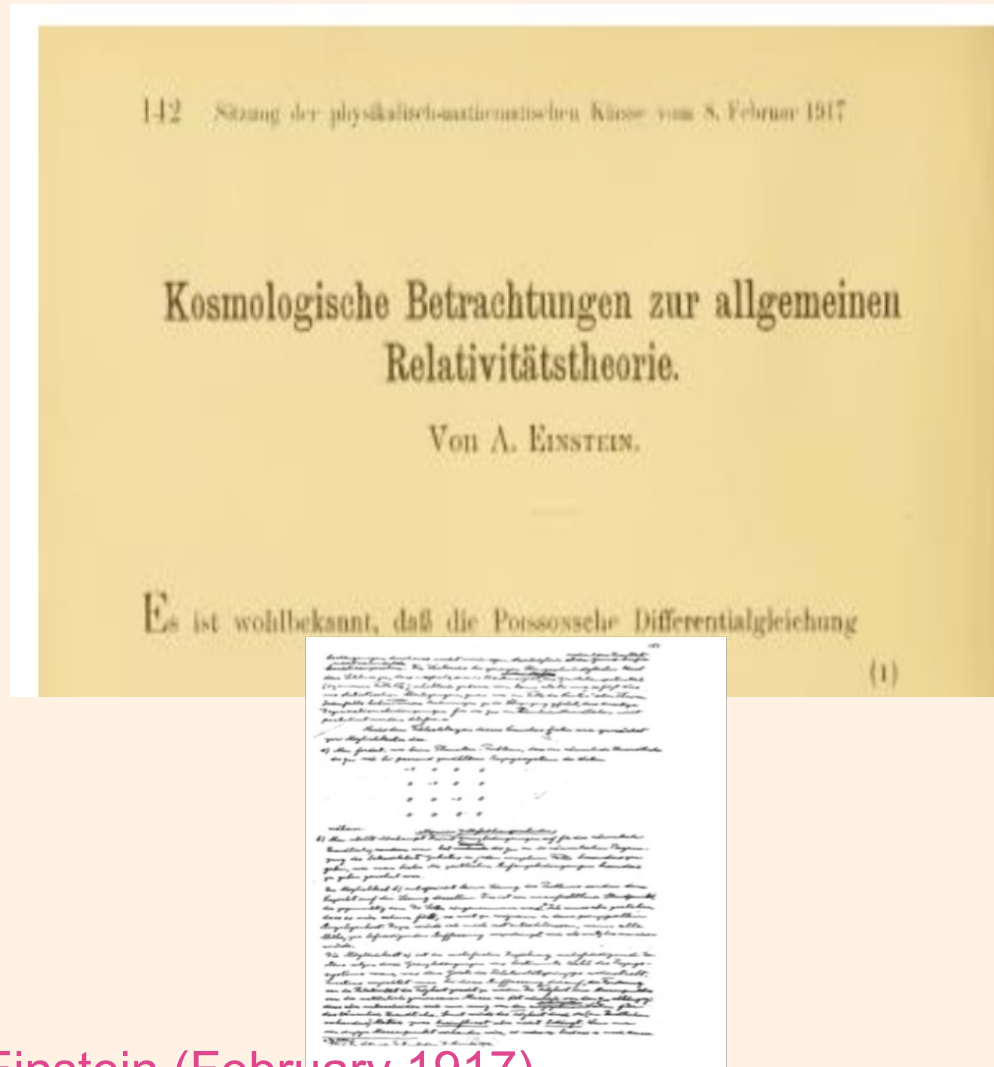
DES dark matter map



# OUTLINE

- A brief history of the Cosmological Constant  $\Lambda$  & Cold Dark Matter
- DESI++ (2025) and DES++ (2026) results:
  - $\Lambda$ CDM or a Dynamical Dark Energy? Try Hyper-Parameters
  - The Future of the Universe
- More with DESI: anomalies, cosmic web
- AI for Cosmology and Cosmology for AI
- The social value of Dark Energy

# EINSTEIN 1917 COSMOLOGICAL CONSTANT



Modified Newtonian

$$\nabla^2 \phi - \lambda \phi = 4\pi\kappa\rho$$

Modified GR

$$G_{\mu\nu} - \lambda g_{\mu\nu} = -\kappa \left( T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)$$

In a static universe:

$$\lambda = \frac{\kappa\rho}{2} = \frac{1}{R^2}.$$

Einstein (February 1917)

English translation: <http://einsteinpapers.press.princeton.edu/vol6-trans/433?ajax>

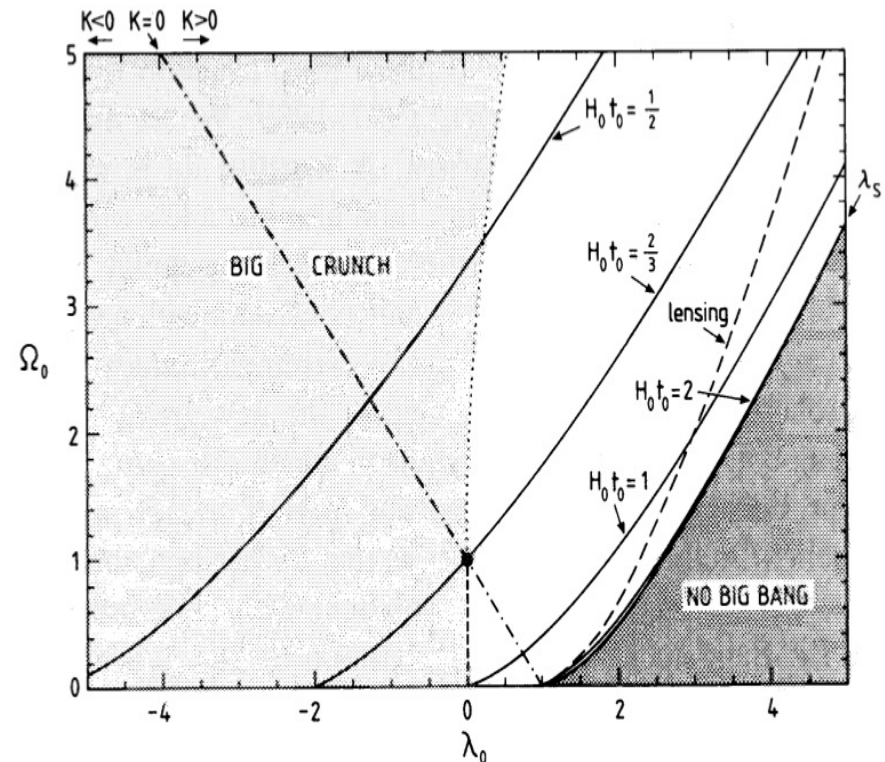
# EVIDENCE FOR $\Lambda$ PRE-1998-99 SNE

- Peebles (1984)
- Weinberg (1989)
- Efsthathiou et al. APM (1990)

$$\Omega_{\Lambda} = 1 - \Omega_m$$

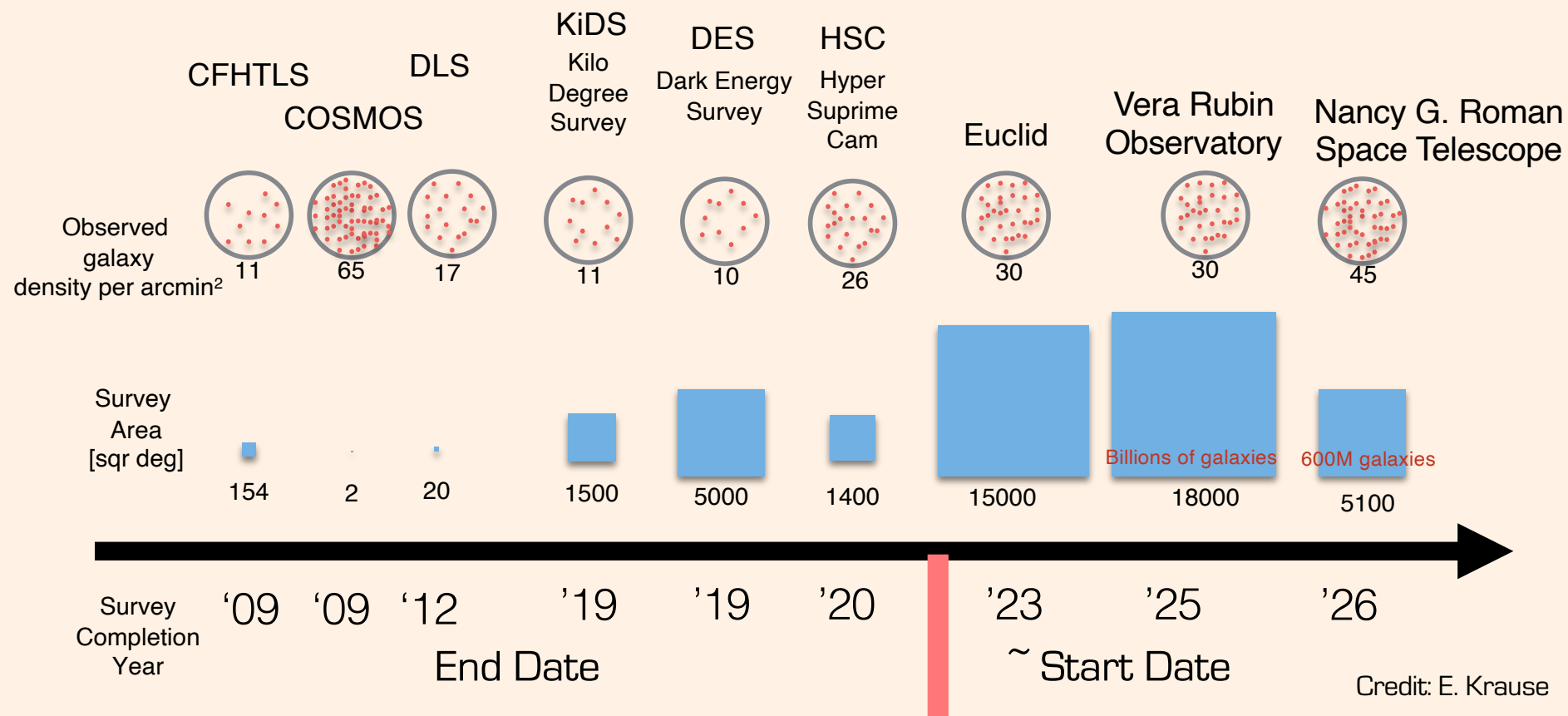
$$= 1 - 0.2 = 0.8$$

OL, Lilje, Primack & Rees (1991)  
Review: Calder & OL (2008, 2010)



**Figure 1.** The phase-space of the density parameter  $\Omega_0$  and the cosmological constant  $\lambda_0 \equiv \Lambda/(3H_0^2)$  with various fundamental constraints. The dashed-dotted line indicates an inflationary (i.e. flat) universe. Note that some open models will have a Big Crunch, while some closed models will expand forever. The solid lines show 4 values for the age of the universe  $H_0 t_0$ , and the dashed line is the constraint of Gott *et al.* (1989) from a normally lensed quasar at  $z=3.27$ . The boundary ( $\lambda_s$ ) of the shaded 'No Big Bang' region corresponds to a coasting phase in the past, while the boundary of the 'Big Crunch' (for  $\Omega_0 > 1$ ) region corresponds to a coasting phase in the future. We see that the permitted range in the  $(\lambda_0 - \Omega_0)$  phase-space is fairly small, but allows values different from the popular point ( $\Omega_0 = 1, \lambda_0 = 0$ ).

# Landscape of Galaxy Surveys: Imaging



# Landscape of Galaxy Surveys: Spectroscopy

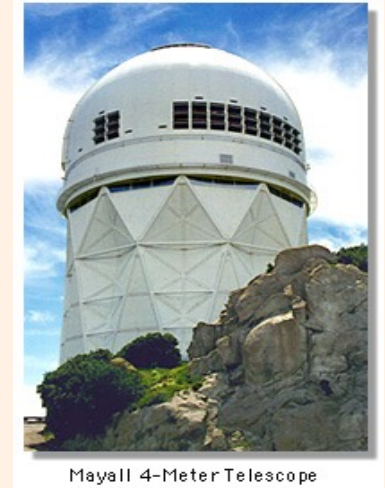


Credit: Hironao Miyatake, SDSS, NOIRLab, NAOJ, ESA/C. Carreau, NASA



# THE TALE OF TWO SURVEYS:

## DARK ENERGY SURVEY (DES) & DARK ENERGY SPECTROSCOPIC INSTRUMENT (DESI)



- Modern instruments on old twin 4m telescopes: DES/DECam (imaging) on Blanco (Chile) and DESI (spectroscopy) on the Mayall (Kitt Peak)

- DES Fermilab-led; DESI LBL-led & international partners

- UCL built both optical correctors

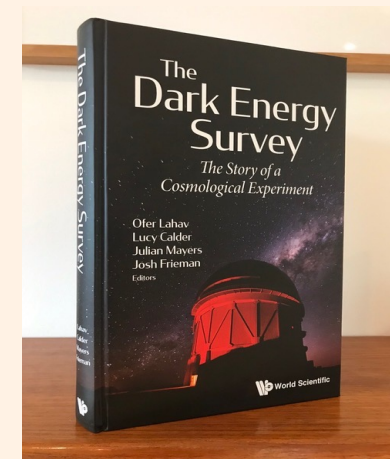
- DES completed 6 seasons in 2019

300M galaxy images (140M with shape measurements)

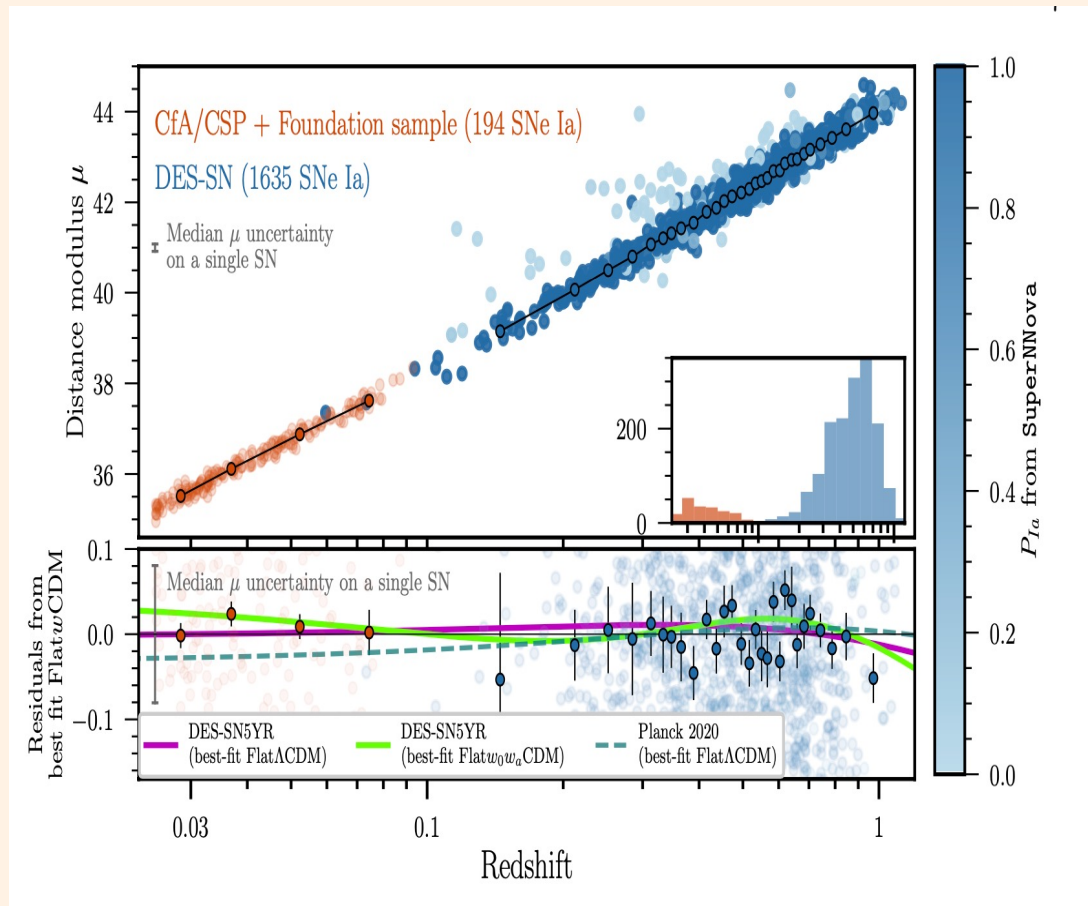
and thousands of SN

- DESI started observations in 2020

47M galaxy+qso spectra



# RESULTS FROM 1500 DES SUPERNOVAE IA (YR5, JAN 2024)



Understanding SN systematics is crucial

Efstathiou (2024)  
Vincenzi & DES (2025)  
Popovic & DES (2025)

arXiv:2401.02929

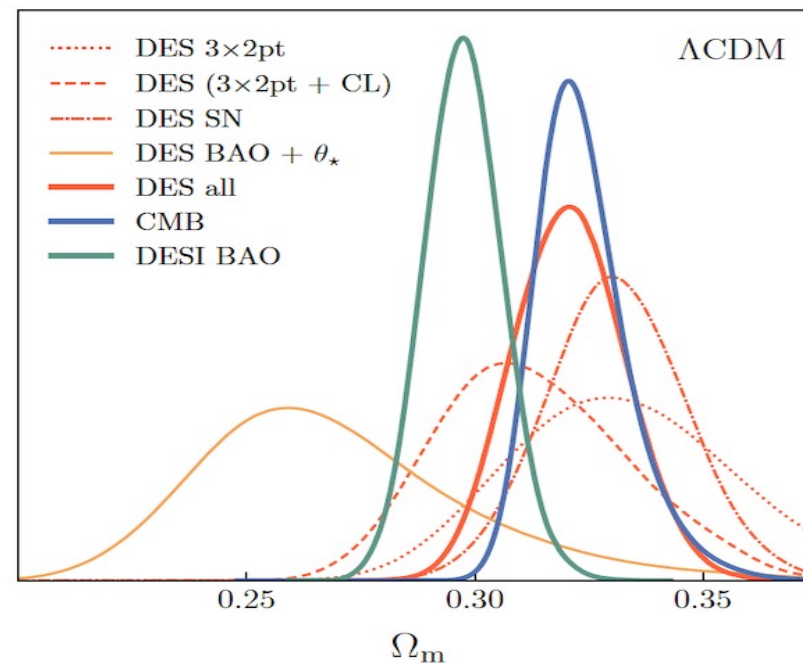
From DES SN+ 3x2pt + SDSS BAO + Planck

$$(\Omega_M, w) = (0.321 \pm 0.007, -0.941 \pm 0.026).$$

DESI (2024, 2025), DES (2026)

$\Lambda$ CDM FITS WELL EACH OF  
BAO, CMB AND SN,  
BUT WITH SLIGHTLY DIFFERENT PARAMETERS, EPS  $\Omega_M$ .  
IT DOESN'T WORK WELL FOR THE COMBINATION.

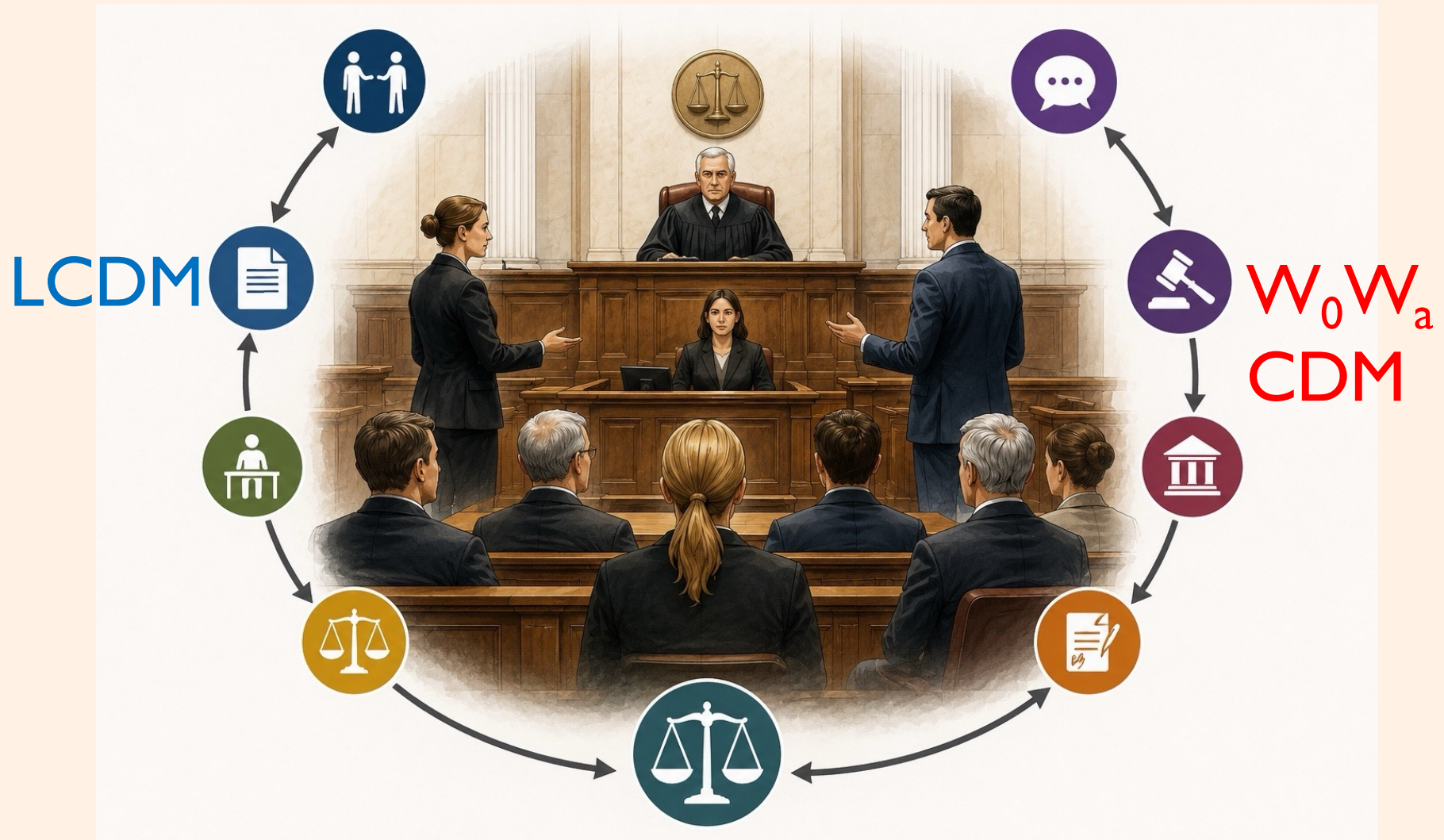
Credit: Claire Lamman



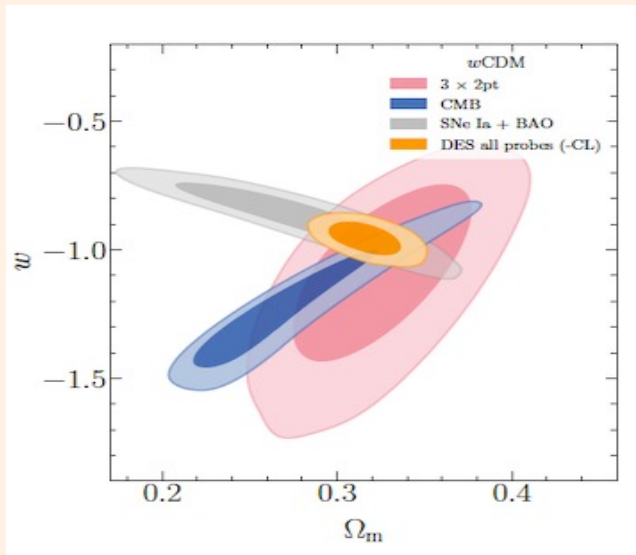
DESY6 2601.14559

See discussion: Eisenstein, Hill, OL, di Valentino & Oldham  
(Nature Astronomy Q&A, Nov 2025)

# COSMOLOGICAL MODELS IN A COURTROOM



## THE CASE FOR LAMBDA ( $w=-1$ EXACTLY) “THE OCCAM’S RAZOR SCENARIO”



DES 2601.14559

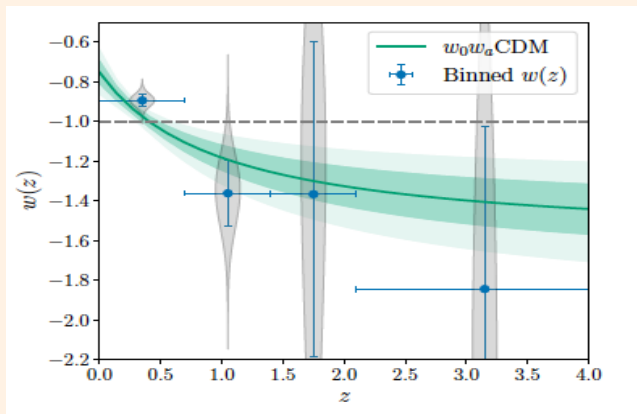
From [DES 3x2pt alone \(2026\)](#) and assuming constant  $w$ CDM

$$w = -1.21^{+0.26}_{-0.20}$$

DES 3x2pt + DESI BAO + SNe + SPT clusters

$$w = -0.98^{+0.021}_{-0.022}$$

DESI for 1<sup>st</sup> bin:  $w=-1$  to within 2 sigma, at around the pivot redshift  $z=0.4$ , where the data are best.



DESI 2503.14738

But note Linder (2007) about CMB driving to  $w=-1$  around  $z=0.4$ .

See also Cortes & Liddle (2024)

# HOW TO RESCUE $\Lambda$ CDM?

## Systematics in SN:

Dovekie (Popovich et al. 2025)

DES SN+ DESI BAO+CMB:

For  $w_0$ waCDM 4.2 sigma  $\rightarrow$  3.2 sigma from  $\Lambda$ CDM.

## CMB:

Sensitivity to Likelihoods and to  
the optical depth of reionization,

e.g. change tau from 0.06 to 0.08, Sailer et al. (2025), cf. JWST

## Curvature:

Non-zero? (Chen & Zaldarriaga 2025)

# HYPER-PARAMETERS FOR MULTI-PROBE DIAGNOSTICS

OL et al. 2000; Hobson, Bridle & OL 2002; Barnel & Peacock 2018;  
Shah & OL (in prep)

Scaling:

$$-2 \log \mathcal{L}_A = A_{\text{SN}} \chi_{\text{SN}}^2 + A_{\text{BAO}} \chi_{\text{BAO}}^2 + A_{\text{CMB}} \chi_{\text{CMB}}^2$$

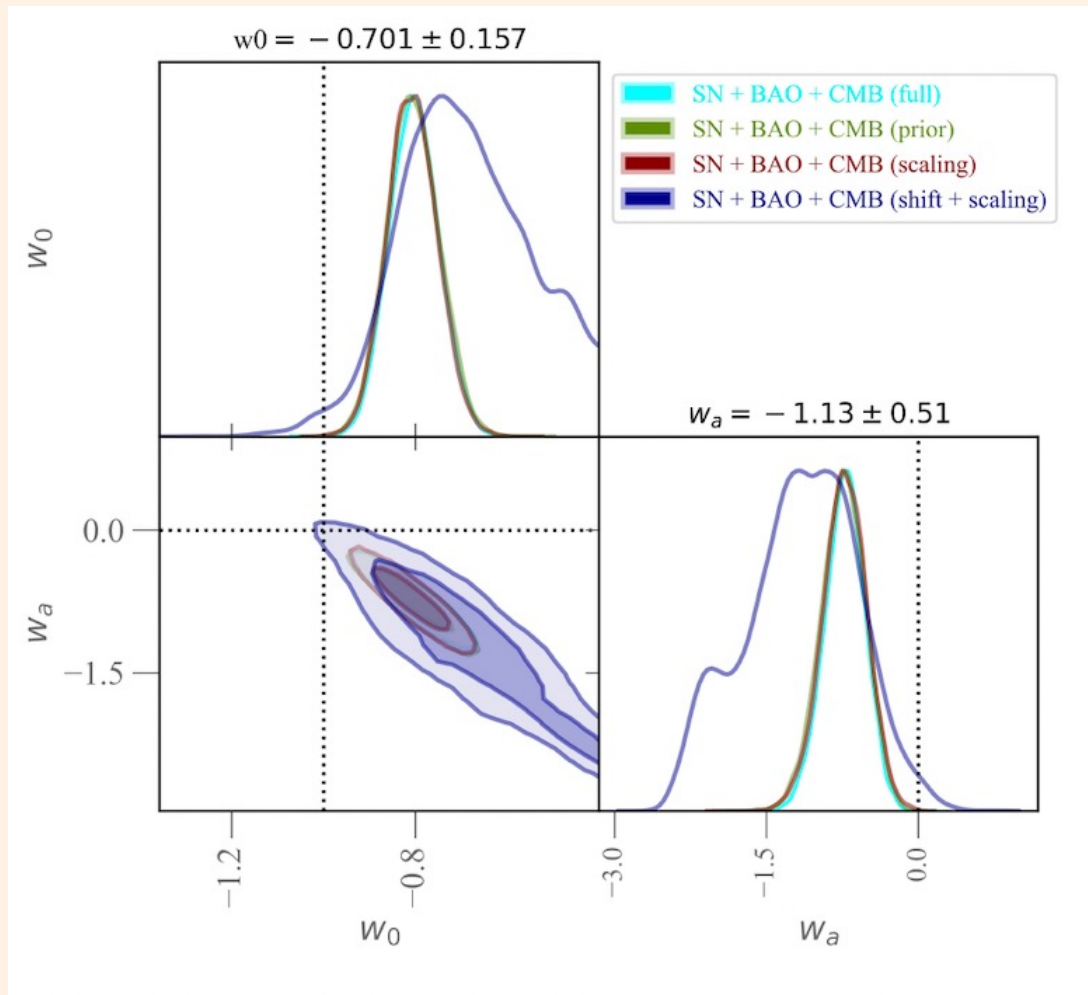
Assume uniform prior on  $\log A$ . (maximum ignorance)

$$-2 \log \mathcal{L}_{\text{Scaling}} = N_{\text{SN}} \log(\chi_{\text{SN}}^2) + N_{\text{BAO}} \log(\chi_{\text{BAO}}^2) + \\ N_{\text{CMB}} \log(\chi_{\text{CMB}}^2) + \text{const.} ,$$

Scaling + Shift:

$$-2 \log \mathcal{L}_B = \sum_j A_j \chi_j^2(\theta + \mathbf{S} \Delta_j) ,$$

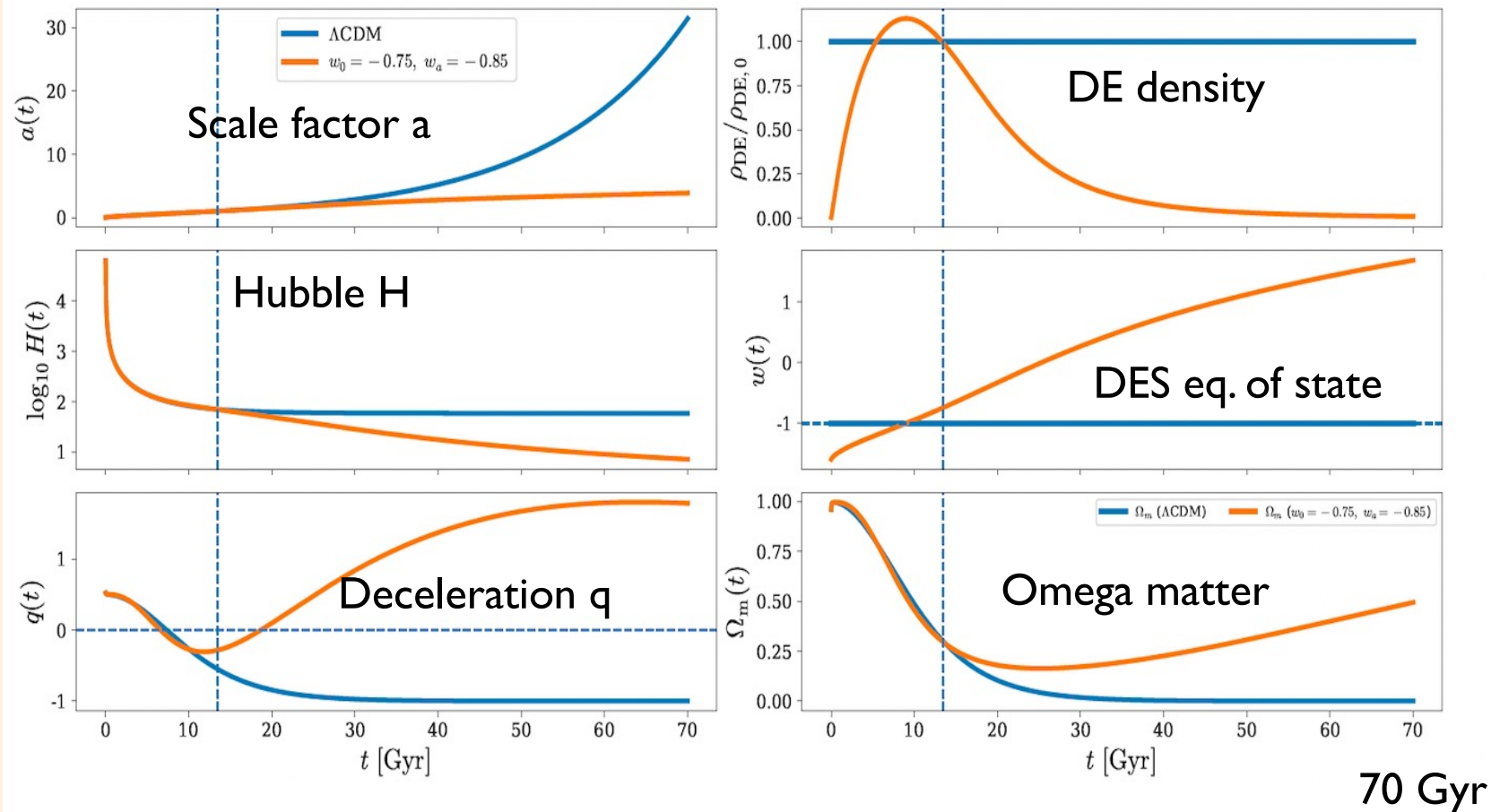
# HYPER-PARAMETERS APPLIED TO BAO+SN+CMB DATA



- Scaling parameters are close to unity,
- eg 0.95 sigma\_SN
- By Evidence ratio, LCDM + HP is weakly preferred to w0waCDM +HP
- Shifts to be understood

Shah & OL (in prep)

# CONSTANT VS. EVOLVING DARK ENERGY: THE FUTURE OF THE UNIVERSE



None of the current proposed  $w_0w_a$  models  
will end in a Big Crunch!

OL & Shah (submitted)

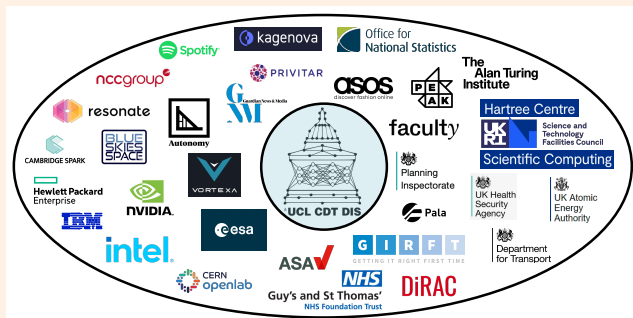
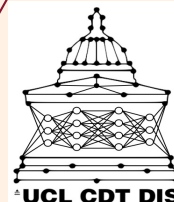
# UCL CENTRE FOR DOCTORAL TRAINING (CDT) IN DATA INTENSIVE SCIENCE (DIS)



Running since 2017  
95 CDT PhD students in 9 cohorts,



80+ academic  
supervisors



30+ industry  
partners

Wide coverage  
of STFC programme



# ANOMALY DETECTION IN DESI SPECTRA

using Autoencoders and other ML techniques

Paul Nathan, University College London PhD student (supervisor Ofer Lahav)



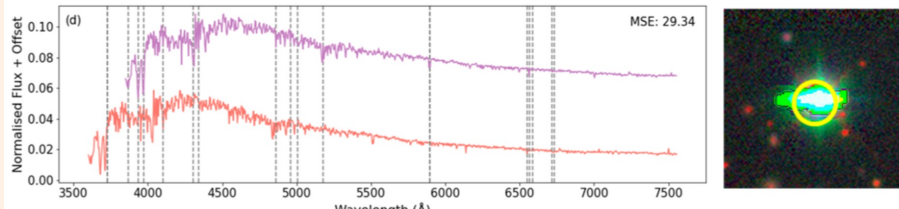
## 1 DESI: Identifying Anomalous Spectra with Variational Autoencoders

Constantina Nicolaou,<sup>1</sup> R.P. Nathan,<sup>1</sup> Ofer Lahav,<sup>1</sup> Antonella Palmese,<sup>2</sup> The DESI Collaboration

<sup>1</sup>University College London, Gower St, London WC1E 6BT, UK

<sup>2</sup>Carnegie Mellon University, 5000 Forbes Ave, Pittsburgh, PA 15213, United States

arXiv:2506.17376



## 2 Finding Pegasus: Leveraging the Manifold from Machine-Learning Dimensionality-Reduction to Enhance Unsupervised Anomaly Detection in DESI Spectra

R. P. Nathan,<sup>1</sup> O. Lahav,<sup>1</sup> and N. Nikolaou<sup>1</sup>

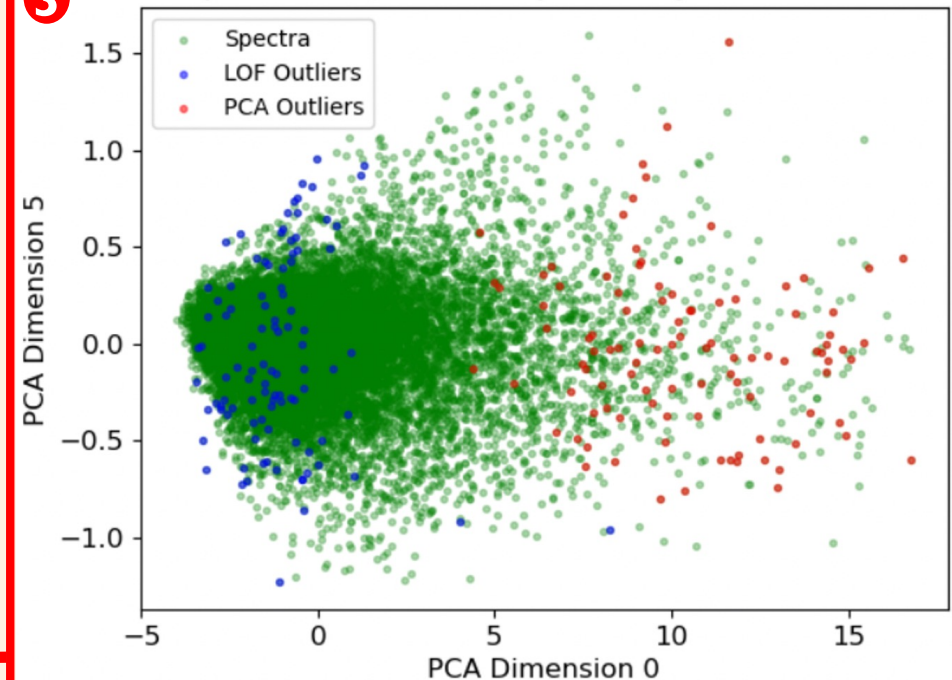
<sup>1</sup>University College London, Gower St, London WC1E 6BT, UK

arXiv:2502.04310

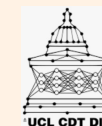


## 3 Starting to apply methods to DESI data

## 3 Spectra FROM QSO SURVEY represented by 6 PCA dimensions



Physics and Astronomy

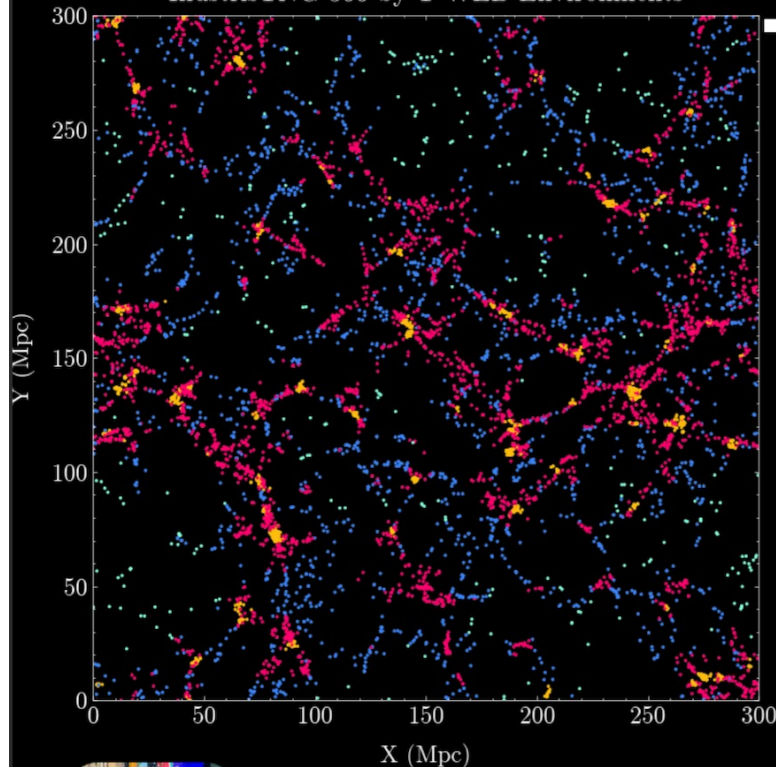


# COSMIC WEB IN DESI WITH AI

## Applying graph learning to real galaxies

Simulation

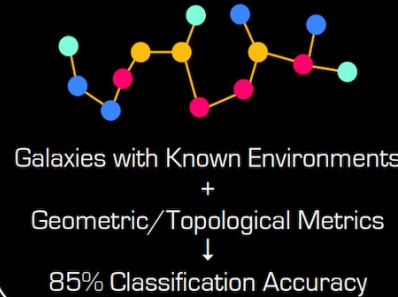
IllustrisTNG-300 by T-WEB Environments



Key: ● Void ● Wall ● Filament ● Cluster

Training

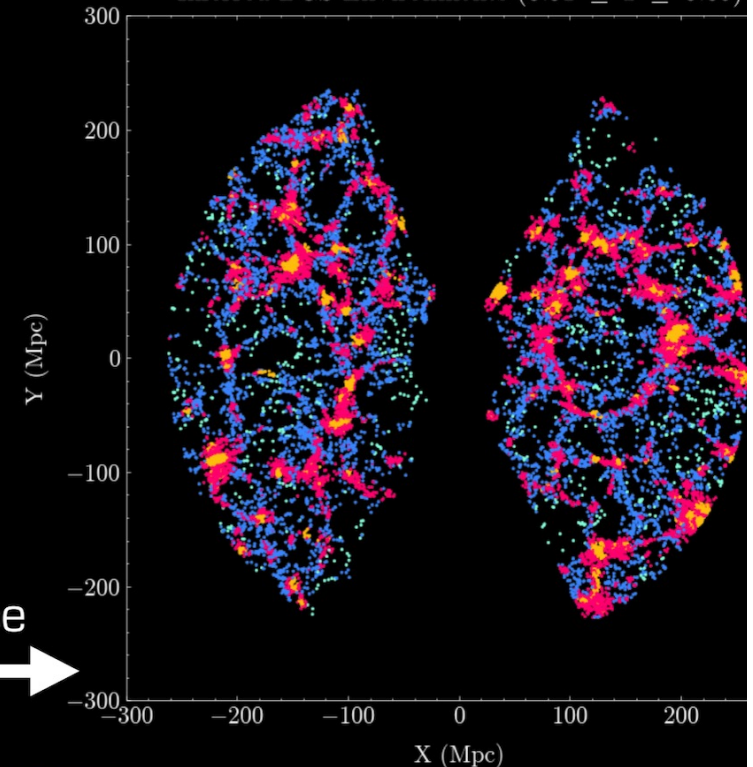
Attentional GNN



Inference

Observation

Inferred BGS Environments ( $0.01 \leq z \leq 0.06$ )



Dakshesh Kololgi

Prof. Ofer Lahav & Prof. Amélie Saintonge

Email: [dakshesh.kololgi.23@ucl.ac.uk](mailto:dakshesh.kololgi.23@ucl.ac.uk)

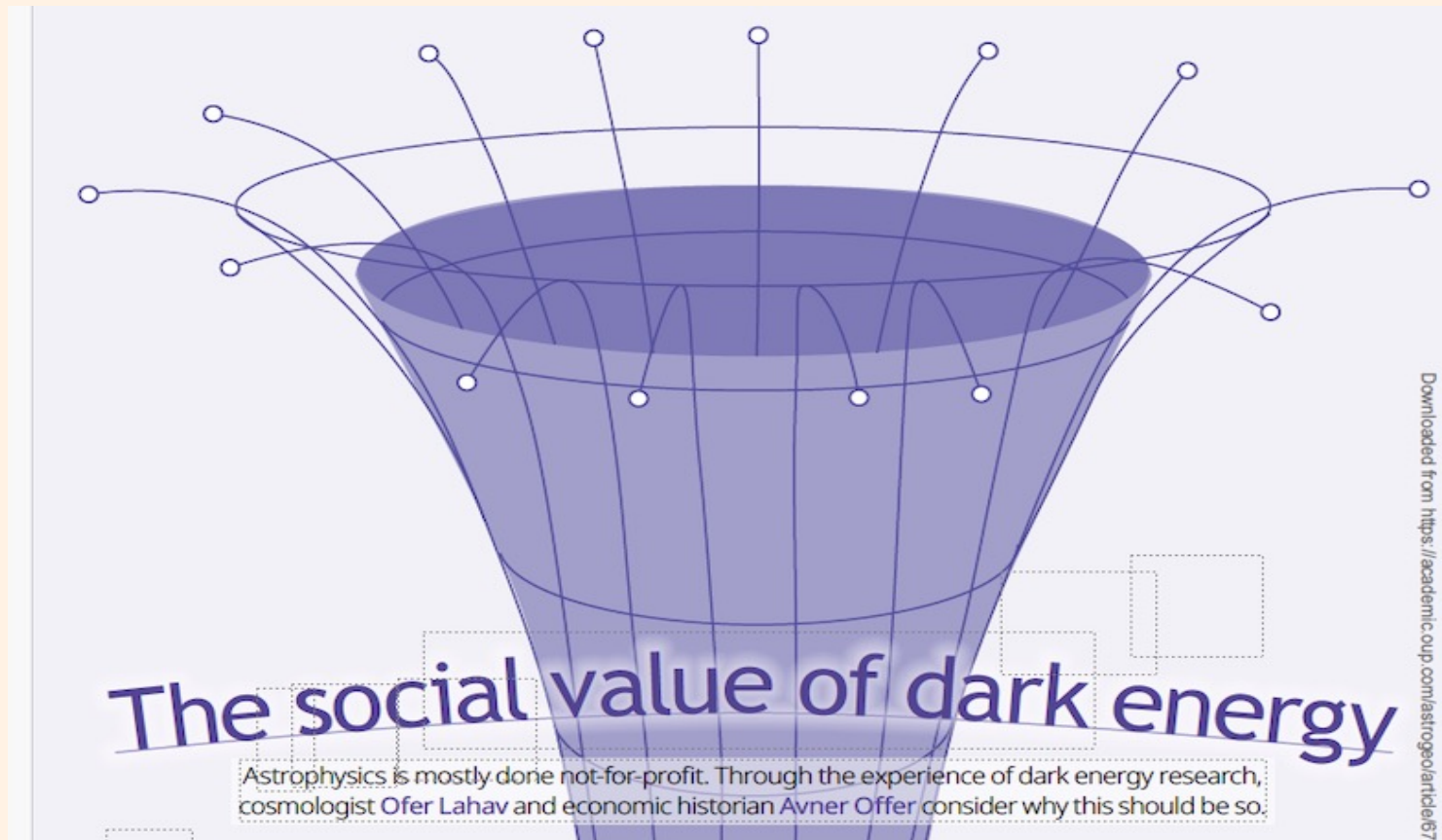
Github: [jdpkololgi](https://github.com/jdpkololgi)



Scan to connect



# WHAT IS THE COST-BENEFIT FOR COSMOLOGY SURVEYS?



OL & Offer, A&G (February 2026), OL & Silk, Nature Astronomy (2021)

# SUMMARY

- 3 decades of '**standard LCDM**': still supported by most current observations - “Not proven guilty”
- **Evolving DE** is emerging – a new paradigm shift? Understand systematics first, e.g. with the aid of Hyper-Parameters.
- **Test LCDM further**: surveys of billions of galaxies are coming online - a golden age for cosmology.
- Could **AI** help us to figure out the universe?

@ANDREW, ANY RANDOM THOUGHTS?



Happy 60<sup>th</sup> Birthday!